Ganpat University
॥ विद्यया समाजोत्कर्षः ॥
Department of Computer Science

B.Sc. Hons. (CA&IT) / Data Science Sem-I

[U11E6CMT] COMPUTER MAINTENANCE AND TROUBLESHOOTING

UNIT-4

Preventative Maintenance and Security

Prepared By:

Pankti Patel

Department of Computer Science Ganpat University

❖ Importance of Routine Maintenance in Computers

Just like we take care of our body (by eating healthy, cleaning, etc.), we must take care of our **computer** to keep it working fast and safely.

Routine maintenance includes:

1. Disk Cleanup

What it means:
It removes unwanted files (like temp files, recycle bin data, cache).

Why it's important:

- Frees up space.
- Makes the computer faster.
- Prevents slow system.

2. Disk Defragmentation

What it means:
It puts scattered files in order on your hard drive.

(When you **save files** on your computer (like photos, videos, documents), they are stored on the hard disk. Over time, these files may get **split into small pieces** and saved in **different places** on the disk. This is called **fragmentation**.)

Why is this a problem?

When you open a file:

- The computer has to **search all over** the disk to collect the pieces.
- This makes your computer **slow**.

What is Disk Defragmentation?

Defragmentation is the process of:

- **Rearranging all the pieces** of a file.
- So that the whole file is stored **together** in one place.

This helps the computer **find and open files faster.**

Why it's important:

- Opens files faster.
- Increases computer speed.
- Reduces hard disk stress.

3. Updates

What it means:
Installing the latest version of Windows or apps.

Why it's important:

- Fixes system errors.
- Protects from viruses.
- Adds new features.

Example:
Like updating your mobile apps for new features and better security.

❖ Conclusion

Routine maintenance keeps your computer:

- Fast
- Safe
- Long-lasting

🛡️ Understanding Malware & Basic Security

1. ⚠️ What is Malware?

Malware = **Malicious Software** (bad software)

Types of Malware:

- **Virus** – Sticks to files and spreads when you open them.(Example: Opening a downloaded game that infects your computer.)

- **Worm** – Spreads on its own through the internet or networks.[It didn't rely on people clicking on email attachments. Instead, it took advantage of a vulnerability (a security hole) in the Windows operating system.

Why it's a Worm: Once a single computer on a network was infected, Worm would automatically scan the network for other vulnerable computers and infect them. It could spread from one company to another across the internet entirely on its own. This is the key feature of a worm: it spreads on its own through networks.]

- **Trojan Horse** – Looks like a safe file or app but is harmful inside.(Example: A "free antivirus" app that actually steals your data.)

often arrived as a seemingly important email, like a shipping invoice, a bank notification, or a job offer. The email would contain a Word document or a link. If you opened the document, it would ask you to "Enable Macros" or "Enable Content" to view it properly.

- **Spyware** – Secretly watches what you do on your computer or phone.(Example: A hidden app tracking your passwords or browsing habits.)

Why it's Spyware: If the executive downloaded and installed the "update," they were actually installing spyware. This software would then secretly watch their activity, stealing sensitive business documents, login credentials, and personal data from their devices. The user had no idea they were being spied on.

- **Ransomware** – Locks your files and asks for money to unlock them.(Example: A message saying "Pay $100 to get your photos back.")

2. 🔐 Basic Security Rules

a) Use Strong Passwords

- Use **uppercase**, **lowercase**, **numbers**, **symbols**
- Password should be 8–12 characters long
- **Example**: Ganpat@2025#IT ✅ (Better than ganpat123 ❌)

b) Avoid Suspicious Links

- Don't click on unknown links or popups
- Check website name carefully
- Download only from trusted sites
- **Example:**
  - ✅ www.bankofindia.com (Real)
  - ❌ www.bank0findia.com (Fake)

3. Why Security is Important

- Keeps your personal data safe
- Protects computer from hackers
- Saves time and avoids damage

🛡️ Antivirus & Anti-Malware Software

1. What is Antivirus?

- A program that **finds and removes viruses**
- Like a **doctor** for your computer

Functions:

- Scans files
- Removes threats
- Protects while browsing or using USB

2. What is Anti-Malware?

- A tool to stop **all types of malware** (not just viruses)
- Stops spyware, ransomware, trojans, etc.

Example:

- **Antivirus** = for simple viruses
- **Anti-malware** = for stronger threats

🌐 Safe Browsing Tips

- Use websites that start with https://
- Don't click on unknown ads or pop-ups
- Don't download cracked software
- Update your browser regularly
- Avoid using free/public Wi-Fi for banking or personal info

📂 Data Protection Strategies

- **Strong Passwords** – Make them long & unique
- **Regular Backups** – Save files to USB or Google Drive
- **Encryption** – Lock important files (need password to open)
- **2-Step Login (MFA)** – Use OTP + password
- **Update Software** – To fix security holes

🛠️ Hands-On Activities

🔧 Part 1: Preventative Maintenance

1. Disk Cleanup

- Press **Windows Key + S**, type **Disk Cleanup**
- Select drive (C:)
- Tick temp files, recycle bin, cache
- Click **OK** → Delete Files
  👉 *Check how much space is cleared*

2. Disk Defragmentation

- Press **Windows Key + S**, type **Defragment and Optimize Drives**
- Select your drive
- Click **Optimize**
  👉 *See if performance improves*

3. Software Update

- Go to **Settings > Update & Security > Windows Update**
- Click **Check for Updates**
  👉 *Updates fix bugs and add security*

🔐 Part 2: Basic Security Tasks

4. Create a Strong Password

- Ask students to create a password with:
    - UPPERCASE + lowercase
    - Numbers
    - Symbols (!, @, #)
    - 8+ characters
      👉 Example: Ganpat@2025#IT
      👉 *Discuss why it is better than ganpat123*

5. Identify Safe vs Fake Links

Show two links:

- ✅ https://www.onlinesbi.com (Real)
- ❌ https://www.onlinesb1.com (Fake)

Ask: Which is safe? Why?

Students write 3 tips to identify safe sites:

- Use HTTPS
- Check spelling
- Trusted sources

6. Antivirus Quick Scan

- Open **Windows Defender** or any antivirus
- Run **Quick Scan**
  👉 *If a threat is found, it will show a message*

7. Safe Browsing Practice

- Open a **safe site** like Wikipedia
- Discuss why to avoid pop-ups or download from unknown sources
  👉 *Always use https:// sites*

8. Backup a File

- Copy a file to **USB / Google Drive**

- Delete the original file
- Restore it from backup
  👉 *Backups help in case of virus or system crash*